Phase 2 — Grade 3: Probability & Statistics

Detailed Revision Guide — full formula breakdowns, edge cases, and common-mistake debugging

1. Basic Probability

Probability measures how likely something is, as a number from 0 (impossible) to 1 (certain).

Formula

```
P(event) = (number of ways the event can happen) / (total number of possible outcomes)
```

Worked Example: bag with 3 red, 2 blue, 5 green (10 total)

```
P(red) = 3/10 = 0.3
P(blue) = 2/10 = 0.2
P(green) = 5/10 = 0.5
```

TIP: Sanity check: probabilities of ALL possible outcomes must sum to 1
0.3 + 0.2 + 0.5 = 1.0. If you list EVERY possible outcome (nothing left out, nothing double-counted), their probabilities always sum to exactly 1 -- something from the list is guaranteed to happen. Use this to catch errors: if your probabilities don't sum to 1 when they should, something was miscounted.

COMMON MISTAKE: Confusing 'the VALUE you're looking for' with 'the COUNT of outcomes that match it'
Real example from practice: for a single fair die, P(die shows 6) was calculated as 6/6 = 1 instead of 1/6. The reasoning error: treating the number '6' itself as if it belonged in the calculation, rather than counting HOW MANY of the six faces actually show a 6. Only ONE face shows a 6 (the other five show 1,2,3,4,5) -- so favorable outcomes = 1, not 6. **Debug habit:** always ask 'how many outcomes, out of the total, actually match what I'm looking for?' -- treat this exactly like the marble-counting example (P(red)=3/10 because 3 marbles ARE red, not because 'red' has some numeric value).

EDGE CASE: An outcome type that repeats counts EACH repeat separately
If a bag had 2 red marbles instead of 1 (e.g. 2 red, 1 blue, 1 green = 4 total), P(red) = 2/4 = 0.5, not 1/4 -- every individual matching item counts toward the numerator, not just 'does this category exist.'

2. The OR Rule and the AND Rule

2.1 OR Rule (Mutually Exclusive Events)

Formula

```
P(A OR B) = P(A) + P(B) -- valid when A and B CANNOT both happen at the same time
```

Worked Example

```
Bag: 3 red, 2 blue, 5 green (10 total)
P(red OR blue) = P(red)+P(blue) = 0.3+0.2 = 0.5
Check by direct counting: 3 red + 2 blue = 5 matching marbles out of 10 = 5/10 = 0.5 ✓
```

EDGE CASE: The OR rule needs a correction when events CAN overlap (not covered in hands-on, but a real gap to know about)

The full, general OR formula is actually: P(A OR B) = P(A) + P(B) - P(A AND B). The '- P(A AND B)' correction removes DOUBLE-COUNTING outcomes that satisfy both A and B at once. In every worked example in this grade (red/blue/green marbles), the events were mutually exclusive (a marble can't be two colors at once), so P(A AND B)=0 and the subtraction term vanished, leaving the simple P(A)+P(B). But if asked e.g. 'P(even number OR number greater than 4)' on a die, 6 satisfies BOTH conditions and would be double-counted unless you subtract P(both). Always check: can both events happen simultaneously? If yes, subtract the overlap.

2.2 AND Rule (Independent Events)

Formula

```
P(A AND B) = P(A) × P(B) -- valid ONLY when A and B are INDEPENDENT
(independent = one event happening has ZERO influence on the other's probability)
```

Full Proof by Enumeration: 2 coin flips

```
ALL possible outcomes: HH, HT, TH, TT (4 total -- because 2 outcomes × 2 outcomes = 4 combos)
P(heads AND heads) = (outcomes matching 'HH') / (total) = 1/4 = 0.25
Matches the formula: P(heads)×P(heads) = 0.5×0.5 = 0.25 ✓
3 coin flips -- ALL outcomes: HHH,HHT,HTH,HTT,THH,THT,TTH,TTT (8 total = 2×2×2)
P(HHH) = 1/8 = 0.125. Matches 0.5×0.5×0.5 = 0.125 ✓
```

TIP: WHY multiply, not add -- the real mechanism

Each additional independent event MULTIPLIES the total number of possible combinations (2 flips = 2×2=4 combos; 3 flips = 2×2×2=8 combos). Multiplying the individual probabilities is just a shortcut for 'count how combinations multiply, then find what fraction matches' -- it is genuinely a counting operation in disguise, not an arbitrary rule.

COMMON MISTAKE: Using P(A)×P(B) for events that are actually DEPENDENT

Real example from practice: given P(badminton)=0.55 and P(cricket)=0.6, it was suggested that P(badminton AND cricket) = 0.55×0.6 = 0.33. But the ACTUAL joint data showed P(badminton AND cricket) = 9/20 = 0.45 -- completely different from 0.33. This mismatch itself proves the multiplication shortcut was invalid here, because badminton and cricket are NOT independent (playing cricket shifts the probability of also playing badminton). **Debug habit:** before multiplying two probabilities together for an AND question, explicitly check independence first (see the test below) -- never multiply by default.

3. Independent vs. Dependent Events

THE TEST to tell them apart

Compare P(Y|X) to the plain/unconditional P(Y):

If P(Y|X) = P(Y) -> INDEPENDENT (knowing X changed nothing about Y)

If P(Y|X) ≠ P(Y) -> DEPENDENT (knowing X shifted the probability of Y)

Worked proof: cricket & badminton (20 students: 12 cricket [9 also badminton], 8 non-cricket [2 badminton])

P(badminton) overall = (9+2)/20 = 11/20 = 0.55

P(badminton | cricket) = 9/12 = 0.75

0.75 ≠ 0.55 -> DEPENDENT (knowing someone plays cricket genuinely raises the badminton probability)

The formula hierarchy (which one is universal, which is a shortcut)

UNIVERSAL (always true, dependent OR independent): P(A and B) = P(A|B) × P(B)

SHORTCUT (only valid when independent): P(A and B) = P(A) × P(B)

Why the shortcut works for independent events: when independent, P(A|B) simplifies down to just P(A) (B gives no new information about A), so the universal formula collapses into the simple shortcut as a special case.

4. Conditional Probability

Formula

P(Y|X) = P(Y and X) / P(X)

In words: out of ALL the cases where X happened, what fraction of THOSE also have Y happening?

The 'shrink the denominator' mechanism, step by step

Classroom: 20 students total. 12 play cricket. Of those 12, 9 also play badminton.

Step 1 -- APPLY the condition ('given cricket'): completely IGNORE the 8 non-cricket students. The entire 'world' shrinks from 20 down to just the 12 cricket players.

Step 2 -- within that shrunk world of 12, count how many ALSO satisfy badminton: 9.

Step 3 -- divide: P(badminton|cricket) = 9/12 = 0.75

Verify this matches the FULL formula (using probabilities over all 20, not just the narrowed group):

P(cricket) = 12/20 = 0.6. P(badminton AND cricket) = 9/20 = 0.45

P(badminton|cricket) = 0.45/0.6 = 0.75 ✓ (same answer -- the direct-count shortcut is just this formula with the common denominator of 20 cancelled out: (9/20)/(12/20) = 9/12)

EDGE CASE: Conditional probability is NOT symmetric: P(Y|X) ≠ P(X|Y) in general
P(badminton|cricket)=0.75 does NOT mean P(cricket|badminton) is also 0.75 -- they are generally DIFFERENT numbers, computed over different narrowed groups (one narrows to cricket players, the other would narrow to badminton players). This exact asymmetry -- knowing P(Y|X) doesn't hand you P(X|Y) for free -- is PRECISELY the problem Bayes' Theorem exists to solve (see Section 5).

EDGE CASE: What if P(X) = 0? (conditioning on something impossible)
The formula P(Y|X)=P(Y and X)/P(X) involves dividing by P(X). If P(X)=0 (the condition itself is impossible), the division is undefined -- conditional probability given an impossible event has no meaningful value. This rarely comes up in practice but is worth knowing as a boundary case.

5. Bayes' Theorem

Bayes' Theorem flips a conditional probability around: if you know P(B|A) but actually need P(A|B) (the REVERSE direction), Bayes' Theorem gets you there. It is not a separate idea from conditional probability -- it is that SAME formula, rearranged.

5.1 Full Derivation (why the substitution works)

Step by step

Start from TWO versions of the conditional probability formula, written both ways:

P(A|B) = P(A and B) / P(B) ...(i)

P(B|A) = P(A and B) / P(A) ...(ii)

Both share the SAME numerator, P(A and B) -- a single, fixed real-world quantity, however you slice it.

Rearrange (ii) to isolate that shared numerator:

P(A and B) = P(B|A) × P(A)

Substitute this into (i)'s numerator (replacing an UNKNOWN quantity with one built from KNOWN pieces):

```
P(A|B) = [P(B|A) × P(A)] / P(B) <-- this is Bayes' Theorem
```

TIP: The REAL reason for the substitution
P(A and B) is usually something you CANNOT measure directly. The substitution replaces it with an equivalent expression built entirely from quantities you DO have data for (P(B|A) and P(A)). It is not algebra for its own sake -- it is specifically solving 'I need something unmeasurable; here's the same thing, rebuilt from measurable pieces.'

5.2 The Law of Total Probability (finding the missing P(B))

Bayes' formula needs P(B) (e.g. P(positive test), overall, across EVERYONE) -- but this is rarely given directly. It must be built from TWO separate sources: people who have the condition and correctly show B, PLUS people who don't have the condition but show B anyway (a false alarm).

Formula

```
P(B) = P(A)×P(B|A) + P(NOT A)×P(B|NOT A)

(split the whole population into the two sub-groups, and add each group's contribution)
```

5.3 Full Worked Example: Medical Test (every number labeled)

Given

```
P(disease) = 0.01 [this is P(A)]

P(positive | disease) = 0.9 [this is P(B|A) -- test accuracy for sick people]

P(positive | NOT disease) = 0.05 [false-positive rate for healthy people]

Asked: P(disease | positive) [this is P(A|B) -- the REVERSED direction from what's given]
```

Step-by-step solution

```
Step 1 -- find P(NOT disease) = 1 - P(disease) = 1 - 0.01 = 0.99

Step 2 -- find P(positive AND disease) = P(disease)×P(positive|disease) = 0.01×0.9 = 0.009

Step 3 -- find P(positive AND NOT disease) = P(NOT disease)×P(positive|NOT disease)

= 0.99×0.05 = 0.0495

Step 4 -- Law of Total Probability: P(positive) = 0.009 + 0.0495 = 0.0585

Step 5 -- Bayes' Theorem: P(disease|positive) = P(positive AND disease)/P(positive) = 0.009/0.0585

= 0.1538 (about 15.4%)
```

COMMON MISTAKE: Confusing P(B|A) directly with P(A and B) -- SKIPPING the multiplication by P(A)
Real example from practice: P(positive AND disease) was initially written as simply '0.95' (copying the given P(positive|disease) value directly), instead of correctly computing P(disease)×P(positive|disease) = 0.02×0.95 = 0.019. P(B|A) and P(A and B) are DIFFERENT quantities -- P(B|A) is a rate WITHIN the A-group only, while P(A and B) is a share of the ENTIRE population. You must multiply by P(A) to convert the first into the second.

COMMON MISTAKE: Skipping the Law of Total Probability step entirely
A common shortcut-error is dividing by P(disease) or by P(positive|disease) directly, instead of properly building P(positive) from BOTH sub-groups (true positives AND false positives) via the Law of Total Probability. Skipping the false-positive contribution from the healthy group will always make the final answer too high and wrong.

5.4 The Base-Rate Problem (why the answer is 'surprisingly low')

Compare Step 2 and Step 3 above: true positives (0.009) vs. false positives (0.0495) -- the false positives are OVER FIVE TIMES bigger, even though the test is 90% accurate for sick people! This happens because the disease is RARE (only 1% of people have it), so the healthy group is HUGE (99%) compared to the sick group (1%). Even a SMALL false-positive rate (5%) applied to that huge healthy group produces more raw false alarms than the true positives generated from the tiny sick group, even with a high 90% detection rate on that small group. **The underlying cause, in one line:** when the thing being tested for is rare, the sheer SIZE of the 'doesn't have it' group means even a small error rate there can swamp the correct detections from the tiny 'has it' group.

EDGE CASE: This 'base rate' issue gets WORSE the rarer the condition is
If the disease were even rarer (say 0.1% instead of 1%), the healthy group would be even larger relative to the sick group, and P(disease|positive) would drop even further for the SAME test accuracy numbers. The rarity of what you're testing for matters just as much as -- sometimes more than -- the test's own accuracy.

6. Distributions, Mean, and Variance

A **distribution** is every possible outcome paired with its probability -- you've already built one without naming it (the bag of marbles: each color paired with its probability).

6.1 Mean (Expected Value)

Formula

```
Mean = ∑ (value × its probability) -- sum this product across EVERY possible outcome
```

Worked Example 1: spinner (2 with p=0.5, 8 with p=0.5)

```
Mean = (2×0.5) + (8×0.5) = 1 + 4 = 5
```

Worked Example 2: fair six-sided die

```
Mean = (1×1/6)+(2×1/6)+(3×1/6)+(4×1/6)+(5×1/6)+(6×1/6) = 21/6 = 3.5
```

TIP: The mean does NOT have to be a possible outcome
You can never actually roll a 3.5 on a die! The mean is a long-run AVERAGE if the process repeated many times -- not a guaranteed single result. This is a genuinely common point of confusion worth remembering explicitly.

6.2 Variance

Formula

$$\text{Variance} = \sum [\ (\text{value} - \text{mean})^2 \times \text{its probability}\ ]$$

Why SQUARE the difference? (1) makes every term positive -- being 2 below the mean and 2 above should both count as 'equally far,' not cancel out. (2) penalizes bigger deviations more heavily than small ones.

Worked Example 1: spinner (2 with p=0.5, 8 with p=0.5), mean=5

$$\text{Outcome 2: } (2-5)^2 \times 0.5 = (-3)^2 \times 0.5 = 9 \times 0.5 = 4.5$$

$$\text{Outcome 8: } (8-5)^2 \times 0.5 = (3)^2 \times 0.5 = 9 \times 0.5 = 4.5$$

$$\text{Variance} = 4.5 + 4.5 = 9$$

Worked Example 2: fair six-sided die, mean=3.5

$$(1-3.5)^2 \times 1/6 = 6.25 \times 0.1667 = 1.0419$$

$$(2-3.5)^2 \times 1/6 = 2.25 \times 0.1667 = 0.3751$$

$$(3-3.5)^2 \times 1/6 = 0.25 \times 0.1667 = 0.0417$$

$$(4-3.5)^2 \times 1/6 = 0.25 \times 0.1667 = 0.0417$$

$$(5-3.5)^2 \times 1/6 = 2.25 \times 0.1667 = 0.3751$$

$$(6-3.5)^2 \times 1/6 = 6.25 \times 0.1667 = 1.0419$$

$$\text{Variance} = \text{sum of all six terms} \approx 2.9167$$

COMMON MISTAKE: Rounding fractions too early (e.g. using 1/6 ≈ 0.16 or 0.17)
1/6 is a REPEATING decimal (0.16666...). Rounding to just 1-2 decimal places (0.16 or 0.17) loses enough precision that, compounded across SIX separate terms in a variance calculation, the final answer drifts noticeably from the true value. **Debug habit:** keep fractions as fractions throughout the calculation where possible, or round to at least 4 decimal places (0.1667) if converting to decimals, and only accept small rounding drift in the FINAL answer, not at every intermediate step.

COMMON MISTAKE: Forgetting to subtract the mean before squaring
A frequent slip is squaring the raw VALUE itself (e.g. $2^2=4$) instead of squaring the DEVIATION from the mean (e.g. $(2-5)^2=9$). Variance is about distance FROM THE MEAN, not the raw value's own size -- always compute (value - mean) FIRST, as its own step, before squaring anything.

EDGE CASE: Standard deviation (a related concept, not yet taught, but commonly confused with variance)
Standard deviation = square root of variance. For the die example, that would be sqrt(2.9167) ≈ 1.708. Standard deviation is in the SAME units as the original data (unlike variance, which is in squared units), which is why it's often reported alongside or instead of variance in real statistics. Flagged here as a natural next question, even though it hasn't been formally covered yet.

7. Full Debugging Checklist for This Grade

1. For any P(X) calculation, am I counting MATCHING outcomes (not confusing the outcome's VALUE with its count)? 2. For an OR question, could both events happen at once? If yes, have I subtracted the overlap? 3. For an AND question, have I explicitly checked independence (compare P(Y|X) to P(Y)) before multiplying? 4. For Bayes' Theorem, have I correctly multiplied P(B|A) by P(A) to get P(A and B), rather than treating P(B|A) itself as the joint probability? 5. Have I used the Law of Total Probability to build the full P(B), including BOTH the true-positive and false-positive contributions? 6. For variance, did I subtract the mean BEFORE squaring, for every single term? 7. Have I avoided rounding repeating fractions (like 1/6) too early, especially across multiple summed terms?